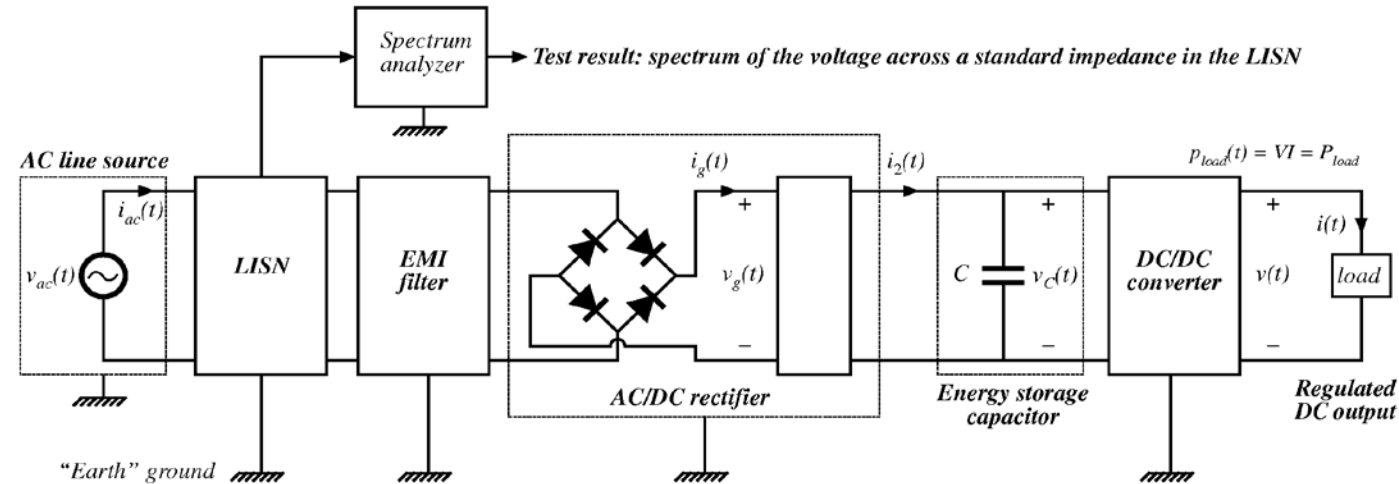


Conducted EMI



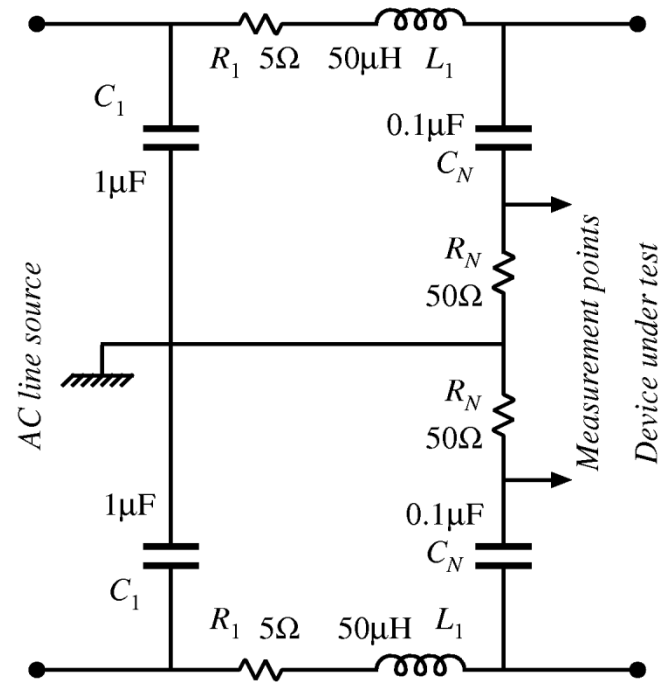
Sample of EMC regulations that include limits on radio-frequency emissions:

European Community Directive on EMC: Euro-Norm EN 55022 or 55081, earlier known as CISPR 22

National standards: VDE (German), FCC (US)

FCC Part 15 B applies to switching power supplies

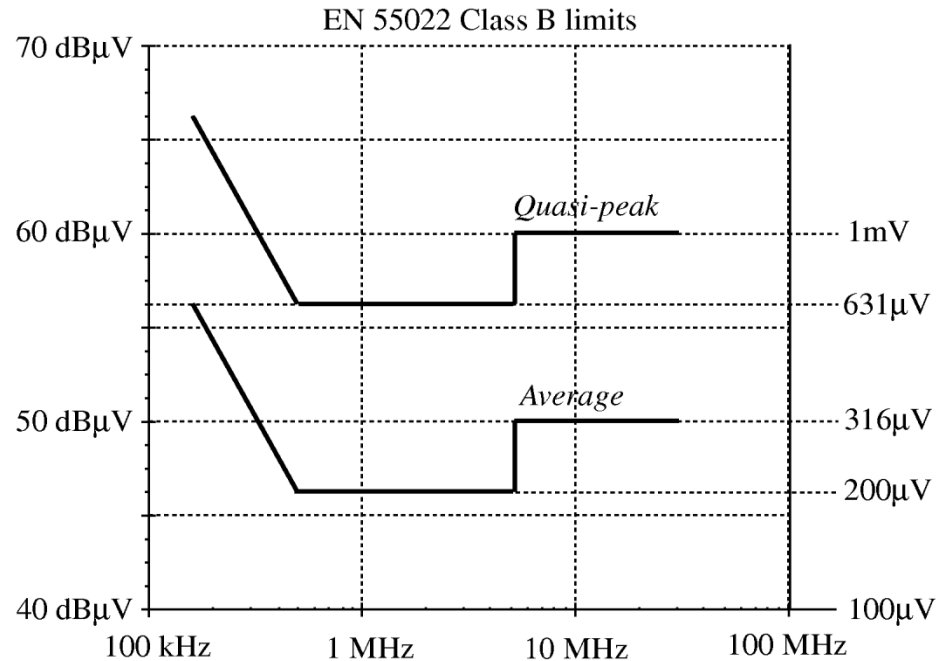
LISN



LISN example

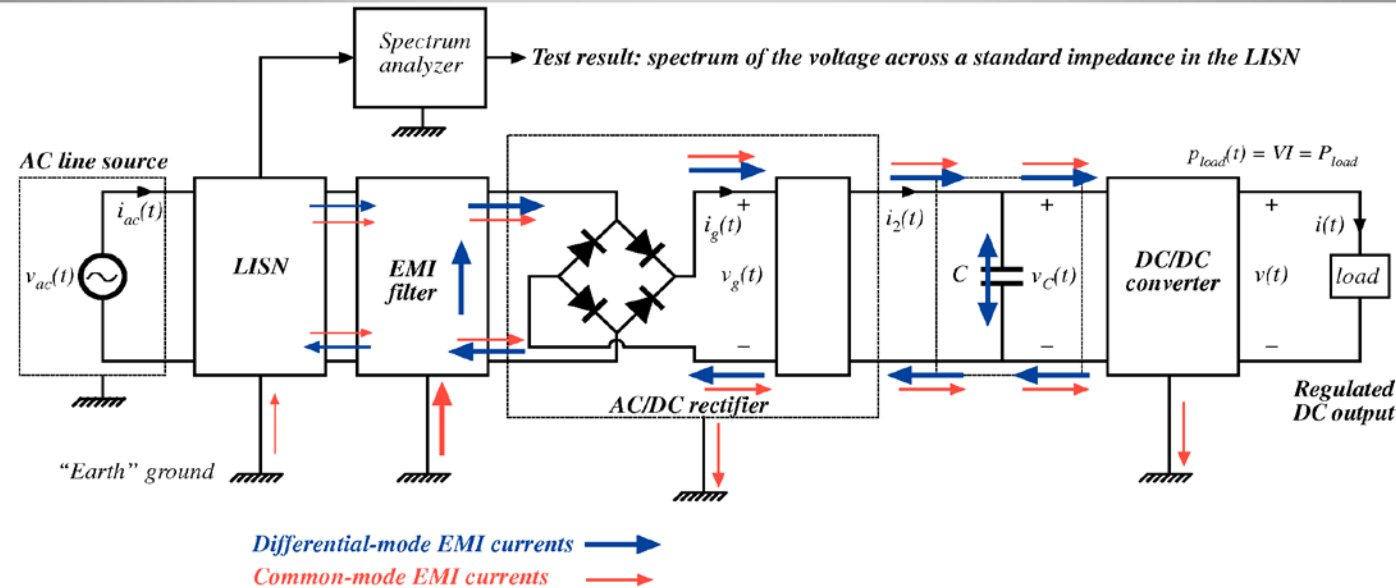
- **LISN:** “Line Impedance Stabilization Network,” or “artificial mains network”
- **Purpose:** to standardize impedance of the power source used to supply the device under test
- Spectrum of conducted emissions is measured across the standard impedance (50 Ω above 150kHz)

An Example of EMI Limits



- Frequency range: 150kHz-30MHz
- Class B: residential environment
- Quasi-peak/Average: two different setups of the measurement device (such as narrow-band voltmeter or spectrum analyzer)
- Measurement bandwidth: 9kHz

Differential and Common-Mode EMI



- **Differential mode EMI:** input current waveform of the PFC. Differential-mode noise depends on the PFC realization and circuit parameters.
- **Common-mode EMI:** currents through parasitic capacitances between high dv/dt points and earth ground (such as from transistor drain to transistor heat sink). Common-mode noise depends on: dv/dt , circuit and mechanical layout.